

$$-V_i + 0.7 + 2V = 0$$

$$V_i = 2.7V$$

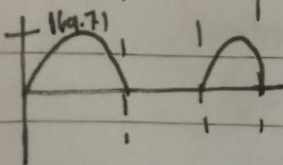
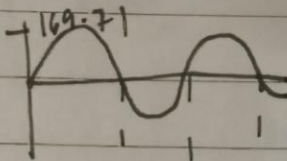
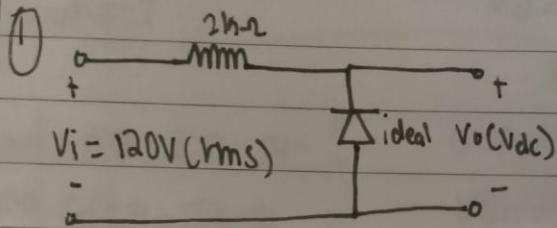
$$r_{ms}(V_2) = V_m \rightarrow \frac{2V_m}{\pi}$$

$$(120)(\sqrt{2}) \quad V_m = 169.71V$$

$$V_i = V_m$$

positive:

$$-V_i + V_o = 0 \quad V_o = V_i$$



negative:

$$V_i = V_o$$

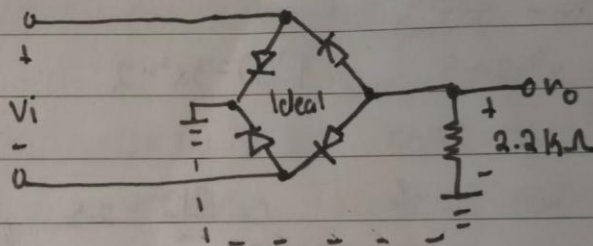
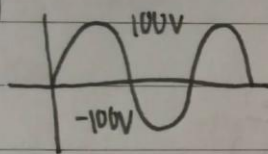
$$V_i + 0 = 0 \quad V_i = 0 \quad \text{or } V_o = 0$$

$$V_{DC} = \frac{V_m}{\pi} = \frac{169.71}{\pi} = 54.02V$$

Date.

Full Wave Rectification

②



Positive: $0 \leq t \leq t/2$

$$-V_i + 0 - V_o + 0 = 0 \quad V_o = -V_i \quad V_o = -100$$

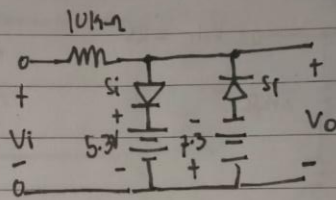
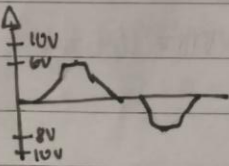
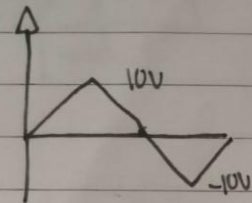
Negative: $t/2 \leq t \leq T$

$$V_i + 0 - V_o + 0 = 0 \quad V_o = V_i \quad V_o = 100$$

Date.

Clippers
Parallel

③



Positive

$$-V_i + 0.7 + 5.3 = 0$$

$$V_i = 6V$$

Negative

$$V_i + 7.3 + 0.7 = 0$$

$$V_i = -8V$$

$$I_A = -10 + 1000(I) + 0.7 + 5.3 = 0$$

$$1000(I) = 10 - 0.7 - 5.3$$

$$I = \frac{4}{10000}$$

$$I = 0.4 \text{ mA}$$

$$-10 + 7.3 + 0.7 + 1000(I) = 0$$

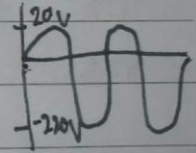
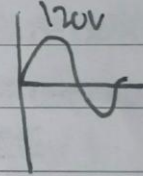
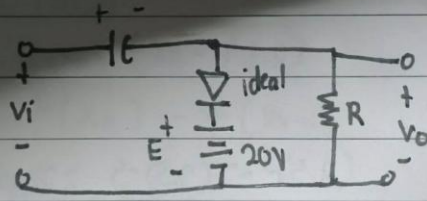
$$1000(I) = 10 - 7.3 - 0.7$$

$$I = 0.2 \text{ mA}$$

+ - + 3V -

Date.

④ Clampers



positive

$$-V_i + V_C + 20V = 0$$

$$-V_i + V_C + 10 = 0$$

$$-120 + V_C + 20V = 0$$

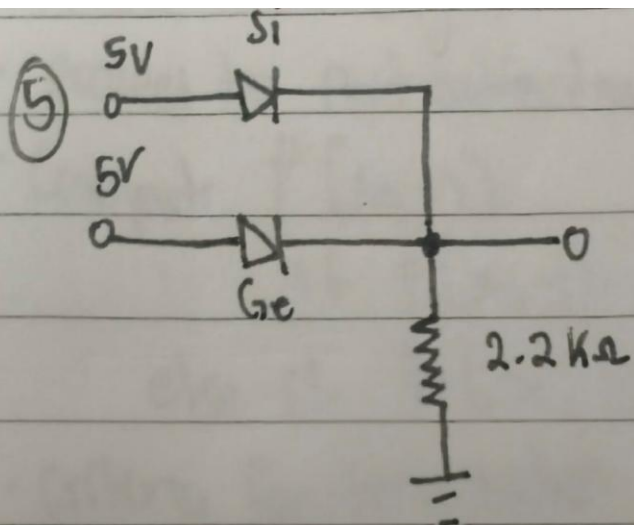
$$V_O = 120V - 100V$$

$$V_O = -120 - 100$$

$$V_C = 100V$$

$$V_O = 20V$$

$$V_O = -220V$$



$$5 - 0.3$$
$$\boxed{= 4.7V}$$